Question			Answer	Marks	AO	Guidance
6	(a)		Explanation, e.g. • The function is positive for all integers so there will be no change of sign • There are two roots between 2 and 3 but the student only checks integers	B1 [1]	2.4	**See appendix** • Or f(2) and f(3) both positive so no change of sign • Or there are two roots between 2 and 3 which will be missed • Must mention integers if they don't explicitly state 2 and 3 • ignore attempts to evaluate f(2) and f(3) • Must have **either** 'no sign change' **or** 'two roots'
	(b)		$\frac{dy}{dx} = 13 - \frac{34}{x} = 0$ $x = \frac{34}{13}$ or $2\frac{8}{13}$	M1 A1 [2]	1.1a 1.1	• Differentiation and setting equal to zero. • Constant must differentiate to zero; sign errors condoned. • Allow for $13 + \frac{k}{x} = 0$ where k is a non-zero constant • Condone poor/no notation • Exact fraction or recurring decimal $2.\dot{6}15384\dot{}$ (if given as a decimal correct dot notation oe must be used) • Ignore y value if seen • isw decimals after correct answer seen

Question 6a

The following comments earn B1
The graph has two roots between 2 and 3 but since the y values and both 2 and 3 are positive, no change in sign will be detected so the student will not find the roots.
The change of sign would go unnoticed between 2 and 3 on the x axis
Both roots are between the positive integers 2 and 3 therefore a change of sign will not be observed.
x=2 , f(x) = 0.933 x = 3, f(x) = 0.147 so there is no change of sign so no roots determined

The following comments earn B0
Student uses integer which when you use f(2) and f(3) both values are positive therefore it could not be used to locate a root **B0 as no explicit mention of change of sign**
As he may skip over the exact moment f(x)=0 due to the interval change/resolution of the spreadsheet data. **B0 as change of sign not mentioned or that there are two roots between 2 and 3**
She uses integer values the roots are between the integers 2 and 3 the graph f(x) is always positive at these values. **B0 as no explicit mention of change of sign or 2 roots**
It has 2 roots. **B0 as no explicit mention of change of sign or 2 roots between integers 2 and 3**
Because there is a ln term in the equation. **B0 as no mention of change of sign or 2 roots**
The student might miss an x value that correlates to f(x)=0 **B0 as no mention of change of sign or 2 roots between 2 and 3.**